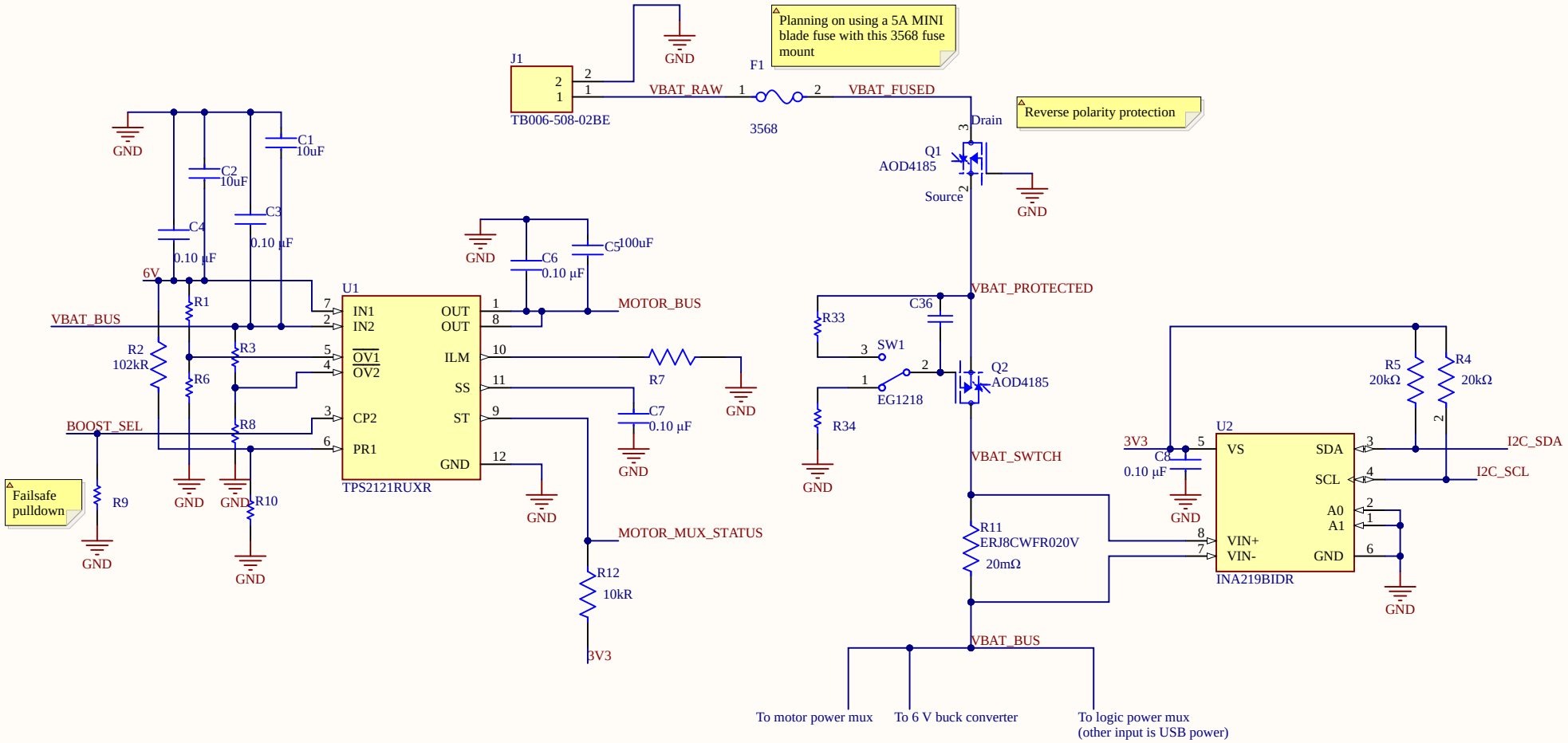
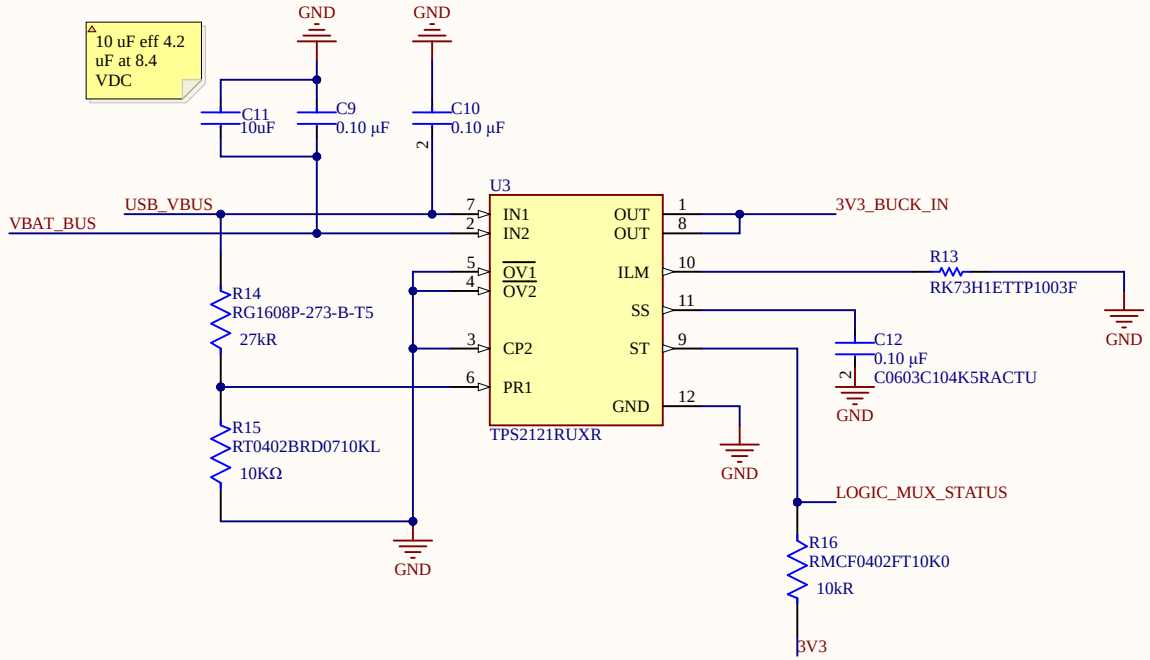
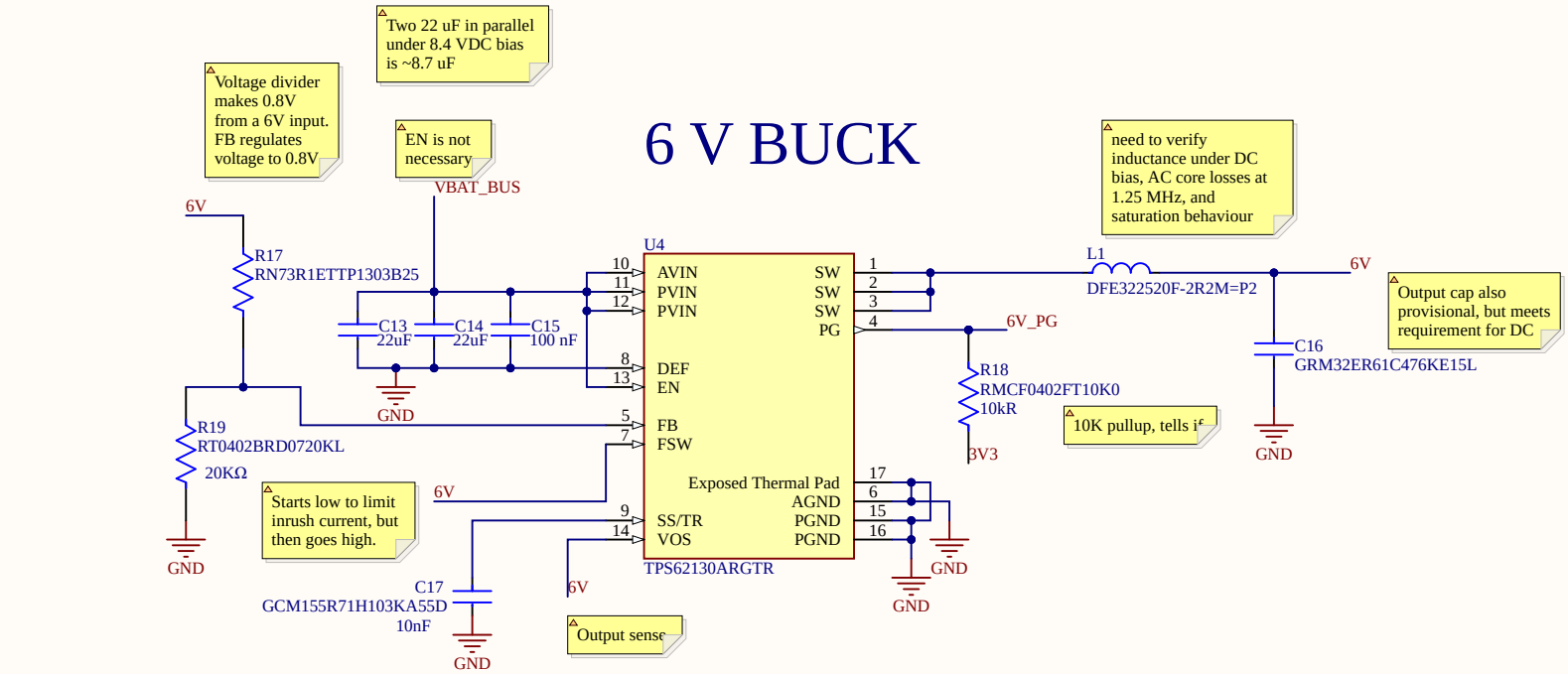
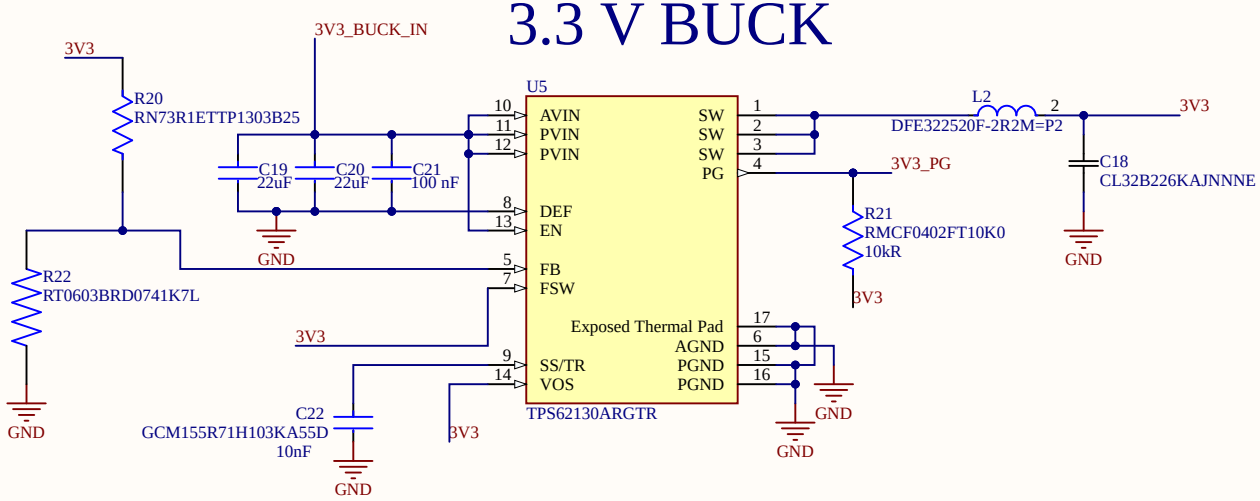



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| Date: 10-16-2026 | | Sheet of |
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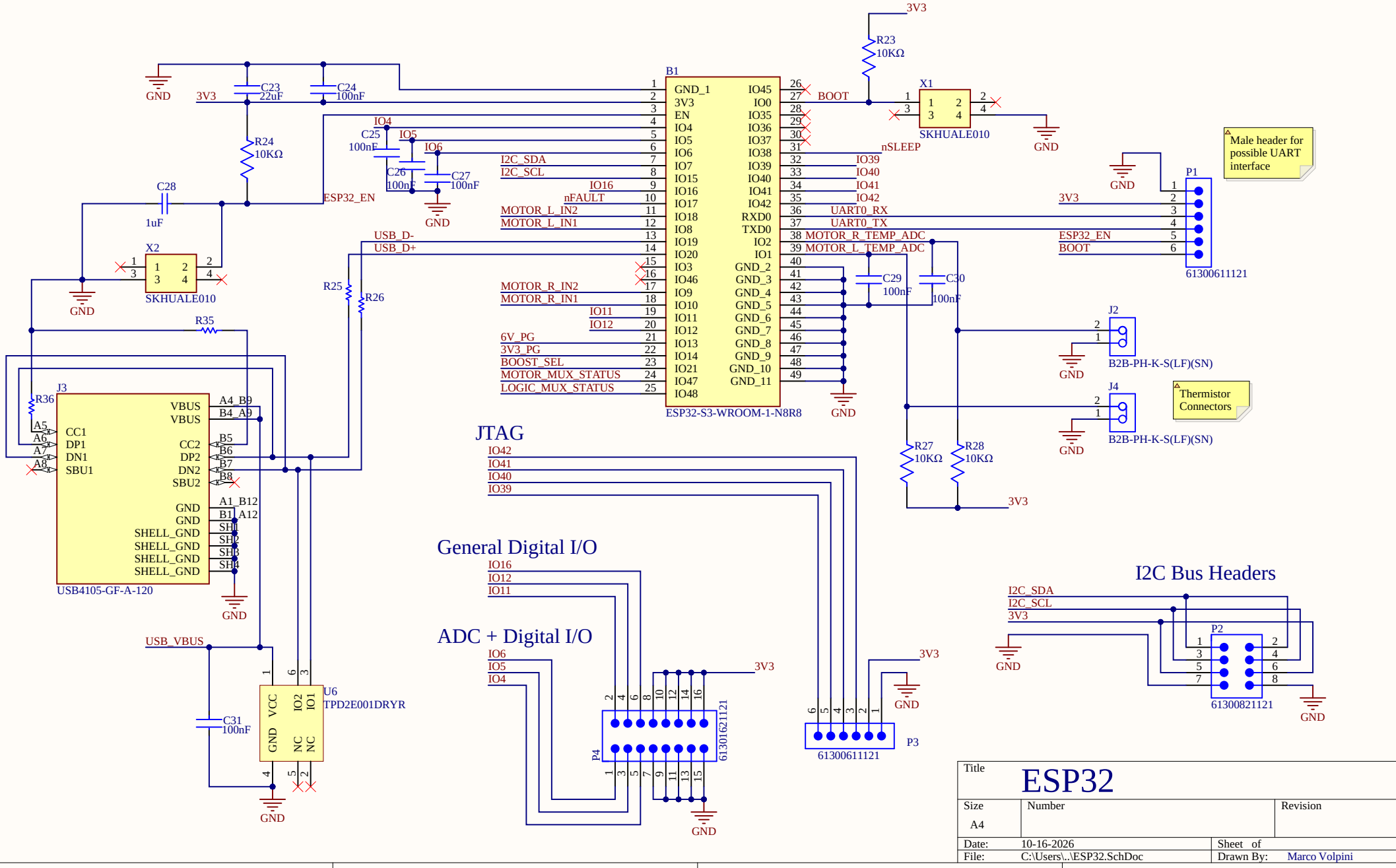



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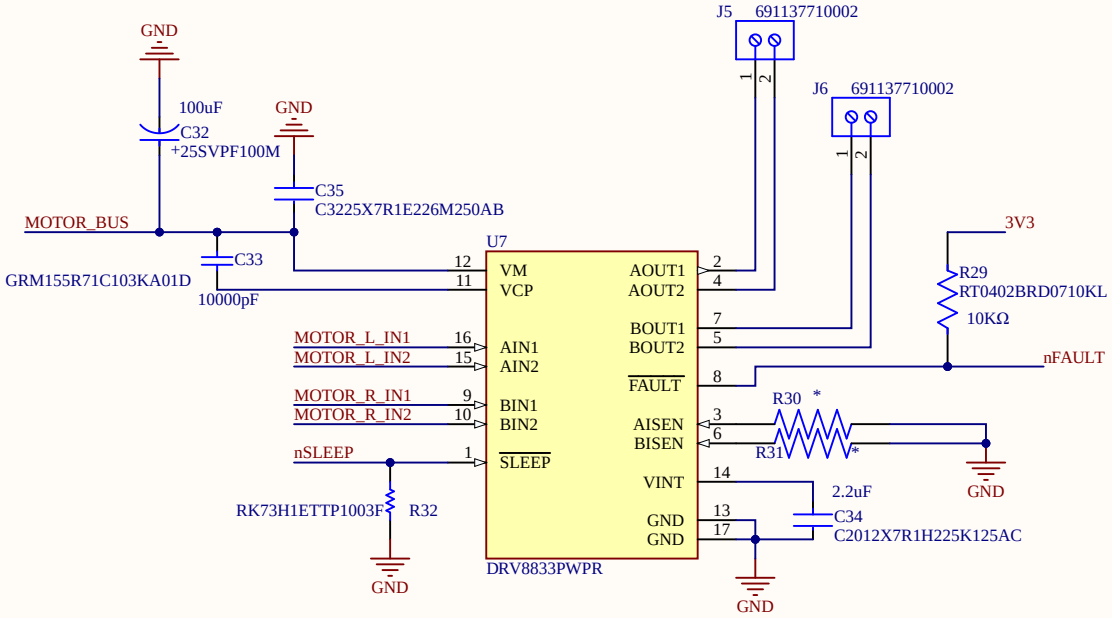


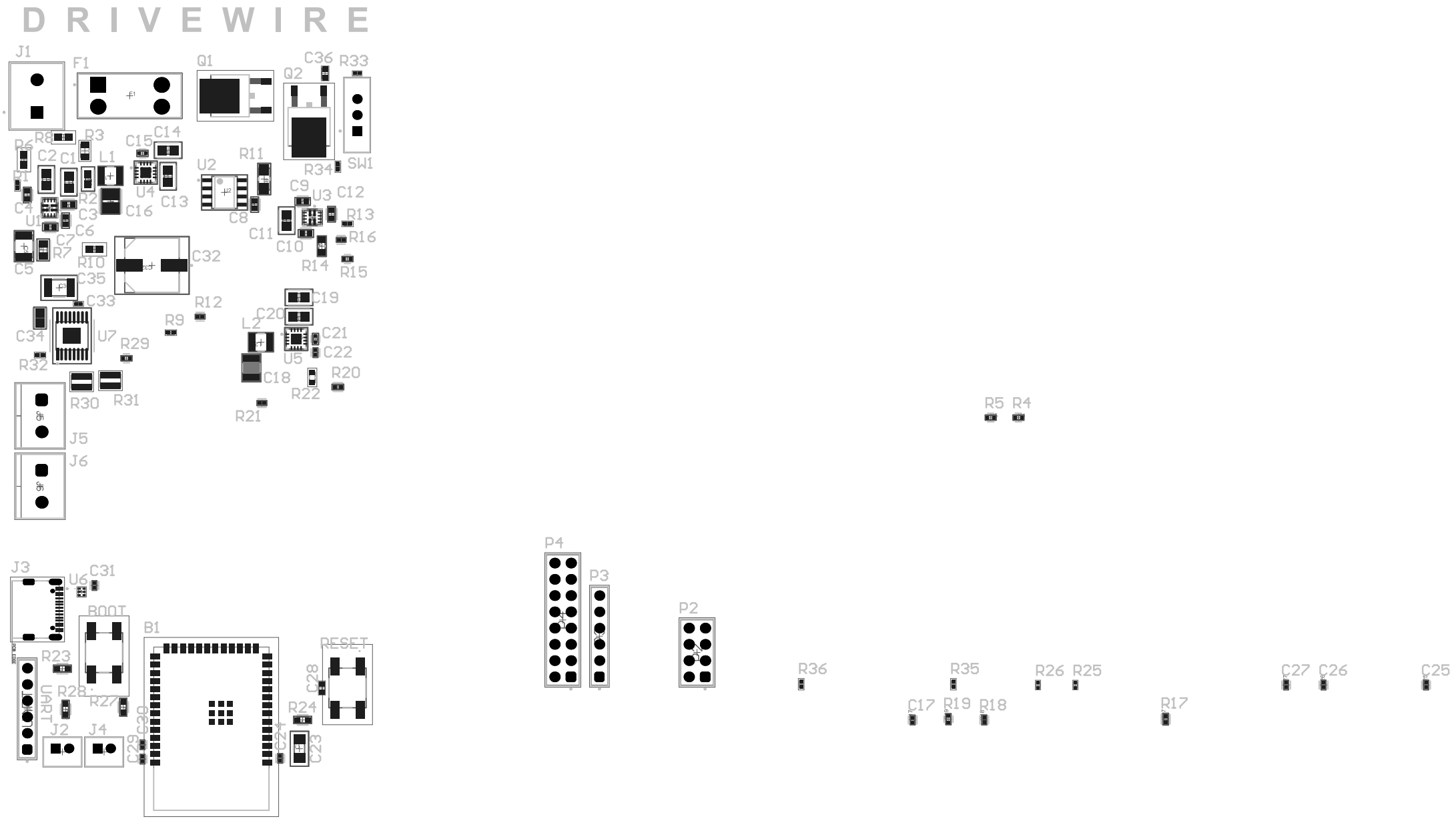

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| Size: A4 | Number: | Revision: |
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|---|---|---|
| Title | ESP32 | |
| Size: A4 | Number | Revision |
| Date: 10-16-2026 | | Sheet of |
| File: C:\Users\..\ESP32.SchDoc | | Drawn By: Marco Volpini |




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| Title | Motor Driver | |
| Size: A4 | Number: | Revision: |
| Date: 10-16-2026 | | Sheet of |
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DRIVEWIRE
J1
F1
Q1
Q2
C36
R33
SW1
R8
R3
C15
C14
R6
C2
C1
L1
U2
R11
R34
U4
C13
C9
U3
C12
C4
C16
C8
R13
C3
C11
C10
R16
U1
C6
C7
R14
R7
R10
R15
C5
C32
C35
C33
C19
R12
C20
R9
L2
C21
C34
U7
U5
C22
R29
R32
C18
R20
R30
R31
R22
R21
J5
J6
R5
R4
P4
P3
P2
J3
U6
C31
BOOT
B1
RESET
R23
R36
R35
R26
R25
C27
C26
C25
R28
R27
C28
R24
C17
R19
R18
R17
J2
J4
C29
C30
C24
C23

| Line # | Name | Description | Designator | Revision ID | Revision State | Revision Status | Quantity | Manufacturer 1 | Manufacturer Part Number 1 | Manufacturer Lifecycle 1 | Supplier 1 | Supplier Part Number 1 | Supplier Unit Price 1 | Supplier Subtotal 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | ESP32-S3-WROOM-1-N8R8 | 2.4GHz 802.11 b/g/n/e/i+BT 5 (LE) + Bluetooth Mesh 8MB Flash 8MB PSRAM 15dBi PCB Antenna | A1 | | Not managed | | 1 | | | | Mouser | 356-ESP32SWROOM1N8R8 | | |
| | GRM0188R61H106KE43L | CAP CER 10UF 50V X5R 0603 | C1, C2, C11 | CMP-00335-004402-1 | New From Design | Up to date | 3 | Murata | GRM0188R61H106KE43L | Volume Production | | | | |
| | C0603C104K5RACTU | CAP CER 0.1UF 50V X7R 0603 | C3, C4, C5, C7, C8, C9, C10, C12 | CMP-2036-03061-2 | New From Design | Up to date | 8 | Yageo Group | C0603C104K5RACTU | Volume Production | Mouser | 80-C0603C104K5R | 0.07 | 0.7 |
| | GMK325ABJ107MM-T | CAP CER 100UF 16V 20% X5R 1210 | C5 | CMP-2000-05020-2 | New From Design | Up to date | 1 | TAIYO YUDEN | GMK325ABJ107MM-T | Obsolete | Mouser | 963-GMK325ABJ107MM-T | 2.53 | 2.53 |
| | CL21A226KOQNNNG | CL21 Series 0805 22uF 16V ±10% Tolerance X5R Multilayer Ceramic Chip Capacitor | C13, C14, C19, C20 | CMP-13271-000211-1 | New From Design | Up to date | 4 | Samsung Electro-Mechanics | CL21A226KOQNNNG | Not Recommended for New Design | Mouser | 187-CL21A226KOQNNNG | 0.42 | 1.68 |
| | CC0402KRX7R7BB104 | CAP CER 0.1UF 16V X7R 0402 | C15, C21 | CMP-03422-000048-5 | New From Design | Up to date | 2 | Yageo Group | CC0402KRX7R7BB104 | Volume Production | Newark | CC0402KRX7R7BB104 | 0.055 | 0.05 |
| | GRM32ER61C476KE15L | CAP CER 47UF 16V X5R 1210 | C16 | CMP-2000-06362-2 | Released | Up to date | 1 | Murata | GRM32ER61C476KE15L | Not Recommended for New Design | Digikey | 490-6539-6-ND | 1.43 | 1.43 |
| | GCM155R71H103KA55D | Ceramic Capacitor for Automotive 10nF ±10% 50VDC X7R 0402 Paper T/R | C17, C22 | CMP-2008-03723-3 | New From Design | Up to date | 2 | Murata | GCM155R71H103KA55D | Volume Production | Arrow Electronics | GCM155R71H103KA55D | 0.0145 | 0.145 |
| | CL21B226KAJNNNE | Description | C18 | | Not managed | | 1 | | | | Future Electronics | CL21B226KAJNNNE | | |
| | CL31B223KHHNNNC | Cap Ceramic 22uF 50V X7R ±10% 3040 1206 +125°C Embossed T/R | C23 | CMP-13271-003516-1 | New From Design | Up to date | 1 | Samsung Electro-Mechanics | CL31B223KHHNNNE | Volume Production | Mouser | 187-CL31B223KHHNNNE | 0.5 | 0.5 |
| | CL05B104KP5NNNC | Cap Ceramic 100nF 10V X7R ±10% 0402 +125°C Paper T/R | C24, C25, C26, C27, C29, C30, C31 | CMP-13271-001008-1 | New From Design | Up to date | 7 | Samsung Electro-Mechanics | CL05B104KP5NNNC | Volume Production | Digikey | 1276-1002-1-ND | 0.025 | 0.25 |
| | GRM188R61A105KA61D | Chip Capacitor, 1 µF, +/- 10%, 10 V, 0603 (1608 Metric), RoHS, Tape and Reel | C28 | CMP-2000-06336-1 | Released | Up to date | 1 | Murata | GRM188R61A105KA61D | Not Recommended for New Design | Arrow Electronics | GRM188R61A105KA61D | 0.106 | 0.106 |
| | 2SMPF100M | Conductive Polymer Aluminum Solid Capacitors 100uF 10V (Dia) 100uF ±20% 25V RADIAL SMT SnE 5mm | C32 | CMP-05428-000935-1 | New From Design | Up to date | 1 | Panasonic | 2SMPF100M | Volume Production | Arrow Electronics | 2SMPF100M | 2.42 | 2.42 |
| | GRM155R71C103KA01D | Chip Capacitor, 10 nF, 16 V, 10%, -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel | C33 | CMP-2000-06332-1 | Released | Up to date | 1 | Murata | GRM155R71C103KA01D | Not Recommended for New Design | Arrow Electronics | GRM155R71C103KA01D | 0.0964 | 0.0964 |
| | C2012X7R1H225K125AC | Multilayer Ceramic Capacitors 2.2µF ±10% 50V X7R SMD 0805 | C34 | CMP-2000-06040-2 | Released | Up to date | 1 | TDK | C2012X7R1H225K125AC | Volume Production | Arrow Electronics | C2012X7R1H225K125AC | 0.4917 | 0.4917 |
| | C3225X7R1E226M250AB | Multilayer Ceramic Capacitors 22µF ±20% 25V X7R SMD 1210 | C35 | CMP-08246-001769-2 | New From Design | Up to date | 1 | TDK | C3225X7R1E226M250AB | Volume Production | Arrow Electronics | C3225X7R1E226M250AB | 1.26 | 1.26 |
| | C0603C105K8RACTU | | C36 | | Unknown server | | 1 | | | | | | | |
| | 3568 | | P1 | CMP-2000-05464-2 | Released | Up to date | 1 | Keystone | 3568 | Volume Production | Newark | 3568 | 1.08 | 1.08 |
| | TB006-508-02BE | 2~24 Poles, Screw Type, Horizontal, 5.08 Pitch, 26~14 AWG, Terminal Block Connector | J3 | | Not managed | | 1 | SameSky | TB006-508-02BE | Volume Production | Arrow Electronics | TB006-508-02BE | 0.1024 | 100.04 |
| | B2B-PH-K-S(LF)(SN) | Male Header, Pitch 2 mm, 1 x 2 Position, Height 5 mm, Tail Length 3.4 mm, -25 to 85 degC, RoHS, Bulk | J5, J6 | CMP-2000-05162-1 | Released | Up to date | 2 | JST | B2B-PH-K-S(LF)(SN) | Volume Production | Newark | B2B-PH-K-S(LF)(SN) | 0.094 | 0.188 |
| | USB4105-GF-A-120 | USB/CDMA, 2.0 TYPE-C, HDPT, USP20, SMT, USB Connector Type:USB Type-C, USB Standard:USB 2.0, Gender:Receptacle, No. of Positions:16Pins, Connector Mounting:Surface Mount, Through Hole Mount, Orientation:Right Angle RoHS Compliant: Yes | J8 | | Not managed | | 1 | | | | Newark | USB4105-GF-A-120 | | |
| | 691137710002 | Terminal Block, 5MM Pitch, Horizontal, Pressure Clamp Glow Wire, Pcb, 2Pin, Gy | J9, J6 | CMP-1532-03324-3 | Released | Up to date | 2 | Wurth Elektronik | 691137710002 | Volume Production | Digikey | 732-10955-ND | 0.38 | 0.76 |
| | DFE322520F-2R2M=P2 | DFE322520F-2R2M=P2 Inductor Power Shielded Wirewound 2.2uH 20% 1MHz Metal 0.1A/0.04Ohm DCR 1210 T/R | L2, L3 | CMP-09260-130381681-1 | New From Design | Up to date | 2 | Murata | DFE322520F-2R2M=P2 | Volume Production | Arrow Electronics | DFE322520F-2R2M=P2 | 0.2692 | 0.5384 |
| | 61300611121 | 6x1 Vertical Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 6 pins | P1, P5 | CMP-1532-03070-4 | Released | Up to date | 2 | Wurth Elektronik | 61300611121 | Volume Production | Digikey | 732-5319-ND | 0.33 | 0.66 |
| | 61300821121 | 8x1 Vertical Pin Header WR-PHD, Pitch 2.54 mm, Dual Row, 8 pins | P2 | CMP-1532-03078-4 | Released | Up to date | 1 | Wurth Elektronik | 61300821121 | Volume Production | Digikey | 732-5296-ND | 0.67 | 0.67 |
| | 61306521121 | 6x2 Vertical Pin Header WR-PHD, Pitch 2.54 mm, Dual Row, 16 pins | P4 | CMP-1532-03097-4 | Released | Up to date | 1 | Wurth Elektronik | 61306521121 | Volume Production | Digikey | 732-5299-ND | 1 | 1 |
| | AOD4185 | P-Channel 40V 80A (Tc) 2.5W (Ta), 62.5W (Tc) Surface Mount TO-252, (D-Pak) | Q1, Q2 | | Not managed | | 2 | | | | Digikey | 785-1222-1-ND | | |
| | RK73H1ETFTP5013F | 0.05Ω ±1% 0.1W, 1/10W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thin Film | R1 | CMP-2002-03895-1 | Released | Up to date | 1 | KOA Speer | RK73H1ETFTP5013F | Volume Production | Digikey | 2019-RK73H1ETFTP5013FCT-ND | 0.1 | 0.1 |
| | RMCF0603FG102K | 820kΩ ±1% 0.1W 0603 Thick Film Chip Resistor AEC-Q200 compliant | R2 | CMP-26527-000784-1 | New From Design | Up to date | 1 | Stackpole Electronics | RMCF0603FG102K | Volume Production | Digikey | 738-RMCF0603FG102KCT-ND | 0.1 | 0.1 |
| | RC0805FR-07430KL | | R3 | CMP-2001-01136-1 | New From Design | Up to date | 3 | Yageo Group | RC0805FR-07430KL | Volume Production | Arrow Electronics | RC0805FR-07430KL | 0.042 | 0.084 |
| | RC4527FR-072K2L | | R4, R5 | CMP-2002-07620-2 | Released | Up to date | 2 | Yageo Group | RC4527FR-072K2L | Volume Production | Arrow Electronics | RC4527FR-072K2L | 0.009 | 0.009 |
| | TNPU06031499KB32N00 | | R6, R8, R20 | CMP-2000-03693-1 | New From Design | Up to date | 3 | Vishay | TNPU06031499KB32N00 | Volume Production | Digikey | TNPU6 9KBACT-ND | 2.1 | 6.3 |
| | RN73R4ATE29183B50 | 29.8kΩ ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant Thin Film | R7 | CMP-04015-062785-1 | New From Design | Up to date | 1 | KOA Speer | RN73R4ATE29183B50 | Volume Production | Digikey | 2019-RN73R4ATED29183B50CT-ND | 0.13 | 0.13 |
| | RK73H1ETFP5003F | | R9, R13, R22, R24 | CMP-2002-04173-1 | Released | Up to date | 4 | KOA Speer | RK73H1ETFP5003F | Volume Production | Arrow Electronics | RK73H1ETFP5003F | 0.0154 | 0.154 |
| | ERJ8CWFR020V | | R11 | CMP-2003-03763-2 | Released | Up to date | 1 | Panasonic | ERJ8CWFR020V | Volume Production | Arrow Electronics | ERJ8CWFR020V | 0.4262 | 0.4262 |
| | RMCF0402FT10R0 | 10kΩ ±1% 0.063W 0402 Thick Film Chip Resistor AEC-Q200 compliant | R12, R16, R18, R21 | CMP-26527-000139-1 | New From Design | Up to date | 4 | Stackpole Electronics | RMCF0402FT10R0 | Volume Production | Mouser | 708-RMCF0402FT10R0 | 0.012 | 0.12 |
| | RG1608P-273-B-T5 | Automotive Metal Thin Film Chip Resistor, 0603, 27kΩ, 0.1%, 25ppm/°C, 0.1W, 100V | R14 | CMP-2000-02157-2 | Released | Up to date | 1 | Susumu | RG1608P-273-B-T5 | Volume Production | Arrow Electronics | RG1608P-273-B-T5 | 0.0964 | 0.0964 |
| | RT0402BRD0710KL | RES 10.0K OHM 1/16W 1% SMD 0402 | R15, R29 | CMP-2002-02915-2 | New From Design | Up to date | 2 | Yageo Group | RT0402BRD0710KL | Volume Production | Arrow Electronics | RT0402BRD0710KL | 0.0964 | 0.1928 |
| | ERJ3GEYJ473V | 47kΩ ±5% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thick Film | R17, R20 | CMP-04015-027314-1 | New From Design | Up to date | 2 | KOA Speer | RN73R1ETP13038Z5 | Volume Production | Digikey | 2019-RN73R1ETP13038Z5CT-ND | 0.1 | 0.2 |
| | RT0402BRD072ML | RES 2.2M OHM 0.1% 1/16W 0402 | R19 | CMP-2002-02926-2 | New From Design | Up to date | 1 | Yageo Group | RT0402BRD072ML | Volume Production | Arrow Electronics | RT0402BRD072ML | 0.0964 | 0.0964 |
| | RT0603BRD0741K7L | 41.7kΩ ±0.1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Thin Film | R22 | | Not managed | | 1 | | | | Mouser | 603-RT0603BRD07 41K7L | | |
| | RT0603BRD0710KL | RES 10.0K OHM 1/10W 1% SMD 0603 | R23, R24, R27, R29 | CMP-03412-000618-1 | New From Design | Up to date | 4 | Yageo Group | RT0603BRD0710KL | Volume Production | Arrow Electronics | RT0603BRD0710KL | 0.0964 | 0.3856 |
| | CRCW04022230FKED | | R25, R26 | CMP-2002-03382-1 | Released | Up to date | 2 | Vishay | CRCW04022230FKED | Volume Production | Arrow Electronics | CRCW04022230FKED | 0.0033 | 0.0066 |
| | ERJ8CYJ152V | | R28, R31 | | Not managed | | 0 | | | | Newark | ERJ8CYJ152V | | |
| | RK73H1ETFTP5101F | | R30 | | Unknown server | | 0 | | | | | | | |
| | RC0402FR-072RL | | R36, R56 | | Unknown server | | 0 | | | | | | | |
| | EG1218 | Slide Switch, 600 V, -30 to 70 degC, 3 Pins, RoHS | S1 | CMP-1696-00002-1 | Released | Up to date | 1 | E-Switch | EG1218 | Volume Production | Digikey | EG1903-ND | 0.72 | 0.72 |
| | TPS2121RUXR | PWR MGMT SWTCH/HOUS | U5, U8 | CMP-04610-000009-1 | New From Design | Up to date | 2 | Texas Instruments | TPS2121RUXR | Volume Production | Digikey | 296-53410-6-ND | 2.44 | 4.88 |
| | INA219BIDR | IC MONITOR PWR/CURR/DRIVR 8SOIC | U7 | CMP-0704-00005-3 | Released | Up to date | 1 | Texas Instruments | INA219BIDR | Volume Production | Arrow Electronics | INA219BIDR | 1.88 | 1.88 |
| | TPS2113DRXR | IC MEDIUM CONTROL SOLUTION 8SON | U4, U6 | CMP-04618-000007-1 | New From Design | Up to date | 2 | Texas Instruments | TPS2113DRXR | Volume Production | Newark | TPS2113DRXR | 1.99 | 3.98 |
| | TPD3S501DRYR | Low-Capacitance +/- 15-kV ESD-Protection Array for High-Speed Data Interfaces, 2 Channels, -40 to 85 degC, 6-pin SON (DRY), Green (RoHS and no Sb/Br) | U6 | CMP-0326-00025-3 | Released | Up to date | 1 | Texas Instruments | TPD3S501DRYR | Volume Production | Arrow Electronics | TPD3S501DRYR | 0.3031 | 0.3031 |
| | DRV8833PWPR | IC MOTOR DRIVER 1.5A 16HTSSOP Non-Inv Switches 6.5x6.0x0.5mm 100g/ | U7 | CMP-1529-00004-3 | Released | Up to date | 1 | Texas Instruments | DRV8833PWPR | Volume Production | Mouser | 595-DRV8833PWPR | 2.03 | 3.02 |
| | SHHLHLE010 | | X1, X2 | | Not managed | | 2 | ALPS | SHHLHLE010 | Volume Production | Newark | SHHLHLE010 | 0.3873 | 3.87 |